



STA 2200 PROBABILITY AND STATISTICS II

Purpose By the end of the course the student should be able to solve problems involving probability distributions of a discrete or a continuous random variable.

Objectives

By the end of this course the student should be able to;

- (1) Define the probability mass, density and distribution functions, and to use these to determine expectation, variance, percentiles and mode for a given distribution
- (2) Appreciate the form of the probability mass functions for the binomial, geometric, hyper-geometric and Poisson distributions, and the probability density functions for the uniform, exponential, gamma, beta and normal, functions, and their applications
- (3) Apply the moment generating function and transformation of variable techniques
- (4) Apply the principles of statistical inference for one sample problems.

DESCRIPTION

Random variables: discrete and continuous, probability mass, density and distribution functions, expectation, variance, percentiles and mode. Moments and moment generating function. Moment generating function and transformation Change of variable technique for univariate distribution. Probability distributions: hyper-geometric, binomial, Poisson, uniform, normal, beta and gamma. Statistical inference including one sample normal and t tests.

Pre-Requisites: STA 2100 Probability and Statistics I, SMA 2104 Mathematics for Science

Course Text Books

- 1) RV Hogg, JW McKean & AT Craig *Introduction to Mathematical Statistics*, 6th ed., Prentice Hall, 2003 ISBN 0-13-177698-3
- 2) J Crawshaw & J Chambers *A Concise Course in A-Level statistics, with worked examples*, 3rd ed. Stanley Thornes, 1994 ISBN 0-534- 42362-0

Course Journals:

- 1) Journal of Applied Statistics (J. Appl. Stat.) [0266-4763; 1360-0532]
- 2) Statistics (Statistics) [0233-1888]

Further Reference Text Books And Journals:

- a) HJ Larson *Introduction to Probability Theory and Statistical Inference(Probability and Mathematical Statistics)* 3rd ed., Wiley, 1982
- b) Uppal, S. M. , Odhiambo, R. O. & Humphreys, H. M. *Introduction to Probability and Statistics*. JKUAT Press, 2005
- c) I Miller & M Miller *John E Freund's Mathematical Statistics with Applications*, 7th ed., Pearsons Education, Prentice Hall, New Jersey, 2003 ISBN: 0131246461
- d) Statistical Science (Stat. Sci.) [0883-4237]
- e) Journal of Mathematical Sciences
- f) The Annals of Applied Probability

1. RANDOM VARIABLES

1.1 Introduction

In application of probability, we are often interested in a number associated with the outcome of a random experiment. Such a quantity whose value is determined by the outcome of a random experiment is called a **random variable**. It can also be defined as any quantity or attribute whose value varies from one unit of the population to another.

A **discrete** random variable is function whose range is finite and/or countable, i.e. it can only assume values in a finite or countably infinite set of values. A **continuous** random variable is one that can take any value in an interval of real numbers. (There are *uncountably* many real numbers in an interval of positive length.)

1.2 Discrete Random Variables and Probability Mass Function

Consider the experiment of flipping a fair coin three times. The number of tails that appear is noted as a discrete random variable. X = number of tails that appear in 3 flips of a fair coin. There are 8 possible outcomes of the experiment: namely the sample space consists of

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

$$X = \{0, 1, 1, 2, 1, 2, 2, 3\}$$

are the corresponding values taken by the random variable X .

Now, what are the possible values that X takes on and what are the probabilities of X taking a particular value?

From the above we see that the possible values of X are the 4 values

$$X = \{0, 1, 2, 3\}$$

I.e. the sample space is a disjoint union of the 4 events $\{X = j\}$ for $j=0,1,2,3$

Specifically in our example:

$$\{X = 0\} = \{HHH\} \quad \{X = 1\} = \{HHT, HTH, THH\}$$

$$\{X = 2\} = \{TTH, HTT, THT\} \quad \{X = 3\} = \{TTT\}$$

Since for a fair coin we assume that each element of the sample space is equally likely (with probability $\frac{1}{8}$), we find that the probabilities for the various values of X , called the *probability distribution* of X or the *probability mass function (pmf)*, can be summarized in the following table listing the possible values beside the probability of that value

X	0	1	2	3
$P(X=x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Note: The probability that X takes on the value x , i.e. $p(X = x)$, is defined as the sum of the probabilities of all points in S that are assigned the value x .

We can say that this pmf places mass $\frac{3}{8}$ on the value $X = 2$.

The “masses” (or probabilities) for a pmf should be between 0 and 1.

The total mass (i.e. total probability) must add up to 1.

Definition: The **probability mass function** of a discrete variable is a graph, table, or formula that specifies the proportion (or probabilities) associated with each possible value the random variable can take. The mass function $P(X = x)$ (or just $p(x)$) has the following properties:

$$0 \leq p(x) \leq 1 \text{ and } \sum_{\text{all } x} p(x) = 1$$

More generally, let X have the following properties

- It is a discrete variable that can only assume values $x_1, x_2, \dots, x_n$
- The probabilities associated with these values are $P(X = x_1) = p_1, P(X = x_2) = p_2, \dots, P(X = x_n) = p_n$

Then X is a discrete random variable if $0 \leq p_i \leq 1$ and $\sum_{i=1}^n p_i = 1$

Remark: We denote random variables with capital letters while realized or particular values are denoted by lower case letters.

Example 1

Two tetrahedral dice are rolled together once and the sum of the scores facing down was noted. Find the pmf of the random variable ‘the sum of the scores facing down.’

Solution

+	1	2	3	4
1	2	3	4	5
2	3	4	5	6
3	4	5	6	7
4	5	6	7	8

$$X = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

Therefore the pmf of X is given by the table below

x	2	3	4	5	6	7	8
$P(X=x)$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{3}{16}$	$\frac{1}{4}$	$\frac{3}{16}$	$\frac{1}{8}$	$\frac{1}{16}$

This can also be written as a function

$$P(X = x) = \begin{cases} \frac{x-1}{16} & \text{for } x = 2, 3, 4, 5 \\ \frac{9-x}{16} & \text{for } x = 6, 7, 8 \end{cases}$$

Example 2

The pmf of a discrete random variable W is given by the table below

W	-3	-2	-1	0	1
$P(W=w)$	0.1	0.25	0.3	0.15	d

Find the value of the constant d , $P(-3 \leq w < 0)$, $P(w > -1)$ and $P(-1 < w < 1)$

Solution

$$\sum_{\text{all } w} p(W=w) = 1 \Rightarrow 0.1 + 0.25 + 0.3 + 0.15 + d = 1 \Rightarrow d = 0.2$$

$$P(-3 \leq w < 0) = P(W = -3) + P(W = -2) + P(W = -1) = 0.65$$

$$P(w > -1) = P(w = 0) + P(w = 1) = 0.15 + 0.2 = 0.35$$

$$P(-1 < w < 1) = P(W = 0) = 0.15$$

Example 3

A discrete random variable Y has a pmf given by the table below

Y	0	1	2	3	4
$P(Y=y)$	c	$2c$	$5c$	$10c$	$17c$

Find the value of the constant c hence computes $P(1 \leq Y < 3)$

Solution

$$\sum_{\text{ally}} p(Y=y) = 1 \Rightarrow c(1 + 2 + 5 + 10 + 17) = 1 \Rightarrow c = \frac{1}{35}$$

$$P(1 \leq Y < 3) = P(Y = 1) + P(Y = 2) = \frac{2}{35} + \frac{5}{35} = \frac{1}{5}$$

Exercise 1.1

1. A die is loaded such that the probability of a face showing up is proportional to the face number. Determine the probability of each sample point.
2. Roll a fair die and let X be the square of the score that show up. Write down the probability distribution of X hence compute $P(X < 15)$ and $P(3 \leq X < 30)$
3. Let X be the random variable the number of fours observed when two dice are rolled together once. Show that X is a discrete random variable.
4. The pmf of a discrete random variable X is given by $P(X = x) = kx$ for $x = 1, 2, 3, 4, 5, 6$. Find the value of the constant k , $P(X < 4)$ and $P(3 \leq X < 6)$

5. A fair coin is flip until a head appears. Let N represent the number of tosses required to realize a head. Find the pmf of N
6. A discrete random variable Y has a pmf given by $P(Y = y) = c\left(\frac{3}{4}\right)^y$ for $y = 0, 1, 2, \dots$. Find the value of the constant c and $P(X < 3)$
7. Verify that $f(x) = \frac{2^x}{k(k+1)}$ for $x = 1, 2, \dots, k$ can serve as a pmf of a random variable X .
8. For each of the following determine c so that the function can serve as a pmf of a random variable X .
 - a) $f(x) = cx$ for $x = 1, 2, 3, 4, 5$
 - b) $f(x) = cx^2$ for $x = 0, 1, 2, \dots, k$
 - c) $f(x) = c\left(\frac{1}{6}\right)^x$ for $x = 0, 1, 2, 3, \dots$
 - d) $f(x) = c2^{-x}$ for $x = 0, 1, 2, \dots$
9. A coin is loaded so that heads is three times as likely as the tails. For 3 independent tosses of the coin find the pmf of the total number of heads realized and the probability of realizing at most 2 heads.

1.3 Continuous Random Variables and Probability Density Function

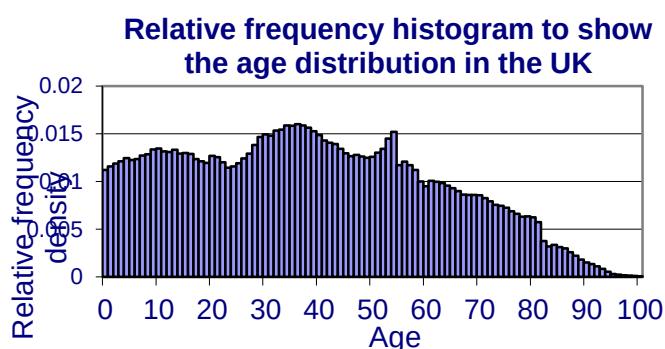
A **continuous** random variable can assume any value in an interval on the real line or in a collection of intervals. The sample space is uncountable. For instance, suppose an experiment involves observing the arrival of cars at a certain period of time along a highway on a particular day. Let T denote the time that lapses before the 1st arrival, then T is a continuous random variable that assumes values in the interval $[0, \infty)$

To illustrate the concept of a probability density function, consider a histogram with unequal class widths. **Frequency density** is plotted on the vertical axis, where

$$\text{frequency density} = \frac{\text{frequency}}{\text{class width}}.$$

One can also draw a **relative frequency histogram** in which the area of a bar corresponds to the **proportion** of the data falling into the corresponding interval. The relative frequency is defined as $\text{Relative frequency} = \frac{\text{frequency density}}{\text{Total frequency}}.$

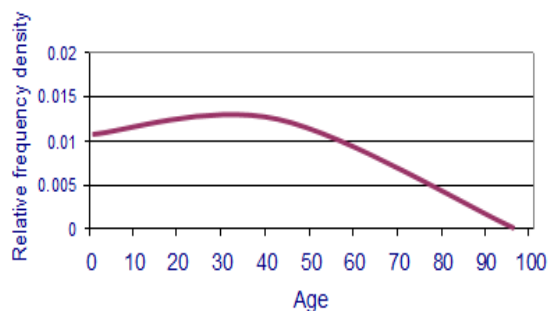
Eg The relative frequency histogram below shows the distribution of ages of the UK population.



The area of each bar corresponds to the proportion of the population with ages in that interval. The **total area** of all the bars is **1**.

The distribution of the ages can be modelled by a curve. This curve is called a **probability density function**.

Definition: A probability density function (or p.d.f.) is a curve that models the shape of the distribution corresponding to a continuous random variable. The function has several important properties. If $f(x)$ is the p.d.f corresponding to a continuous random variable X and if $f(x)$ is defined for $-\infty < X < \infty$ then the following properties must hold:



i) $f(x) \geq 0$ for all x . i.e. the graph of the p.d.f. **never** dips below the **x-axis**.

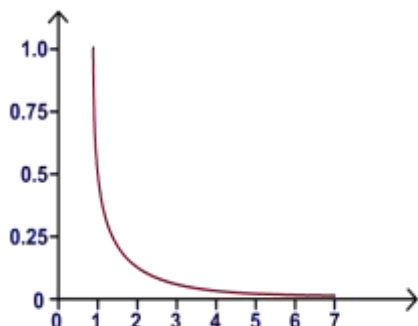
ii) $\int_{-\infty}^{\infty} f(x) dx = 1$ for all x . i.e. the **total area** under a p.d.f. is **1**.

iii) $p(a \leq x \leq b) = \int_a^b f(x) dx$ i.e. **probabilities** correspond to **areas** under the curve.

Example 1: Sketch the graph of each of the following functions. Decide in each case whether it could be the equation of a probability density function: (a) $f(x) = \begin{cases} x^{-3} & X \geq 1 \\ 0 & \text{elsewhere} \end{cases}$

b) $f(x) = \begin{cases} \frac{1}{32}(24x - 3x^2 - 36) & 2 \leq X \leq 6 \\ 0 & \text{elsewhere} \end{cases}$ c) $f(x) = \begin{cases} x^2 - 4x + 3 & 0 \leq X \leq 4 \\ 0 & \text{elsewhere} \end{cases}$

Solution



The function is non-negative everywhere. For f to represent a p.d.f, we need to check that $\int_{\text{all } x} f(x) dx = 1$. But:

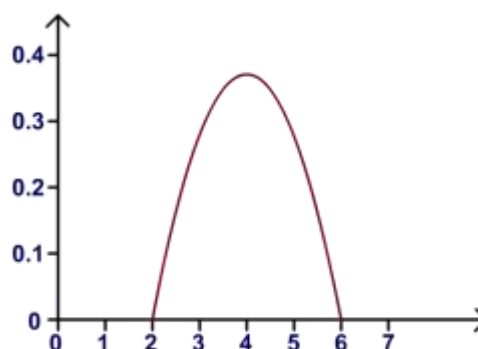
$$\int_{\text{all } x} f(x) dx = \int_1^{\infty} x^{-3} dx = \left[-\frac{1}{2} x^{-2} \right]_1^{\infty} = 0 - \left(-\frac{1}{2} \right) = \frac{1}{2}$$

So $f(x)$ **could not** represent a probability density function.

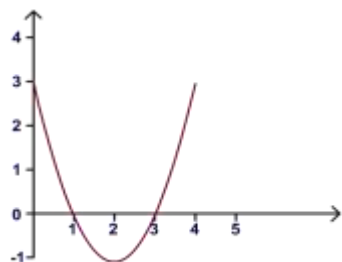
The function is non-negative everywhere.

If f represents a p.d.f., then $\int_{\text{all } x} f(x) dx = 1$

$$\begin{aligned} \int_{\text{all } x} f(x) dx &= \frac{1}{32} \int_2^6 (24x - 3x^2 - 36) dx \\ &= \frac{1}{32} [12x^2 - x^3 - 36x]_2^6 \\ &= \frac{1}{32} [0 - (-32)] = 1 \end{aligned}$$



So $f(x)$ **could** represent a probability density function.



The function is clearly negative for some values of x .

Consequently $f(x)$ cannot represent a probability density function.

Remark A crucial property is that, for any real number x , we have $P(X = x) = 0$ (implying there is no difference between $P(X \leq x)$ and $P(X < x)$); that is it is not possible to talk about the probability of the random variable assuming a particular value. Instead, we talk about the probability of the random variable assuming a value within a given interval which is defined to be the area under the graph of the probability density function between $x = a$ and $x = b$.

Example 2

The time X , in hours, between computer failures is a continuous random variable with density

$$f(x) = \begin{cases} \lambda e^{-0.01x} & \text{for } x > 0 \\ 0, & \text{elsewhere} \end{cases} \quad \text{Find } \lambda \text{ hence compute } P(50 \leq X < 150) \text{ and } P(X < 100)$$

Solution

$$f(x) \geq 0 \text{ for all } x \text{ in } 0 \leq x < \infty \text{ Thus } 1 = \lambda \int_0^{\infty} e^{-0.01x} dx = -100\lambda [e^{-0.01x}]_0^{\infty} = 100\lambda \Rightarrow \lambda = 0.01..$$

$$\text{Now } P(50 \leq X < 150) = 0.01 \int_{50}^{150} e^{-0.01x} dx = [-e^{-0.01x}]_{50}^{150} = e^{-0.5} - e^{-1.5} \approx 0.3834005 \text{ and}$$

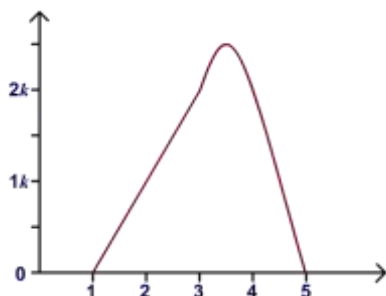
$$P(X < 100) = 0.01 \int_0^{100} e^{-0.01x} dx = [-e^{-0.01x}]_0^{100} = 1 - e^{-1} \approx 0.6321206$$

Example 3 A continuous random variable X is defined by the probability density function

$$f(x) = \begin{cases} k(x-1), & 1 \leq x \leq 3 \\ k(5-x)(x-2) & 3 \leq x \leq 5 \\ 0, & \text{elsewhere} \end{cases}$$

- Sketch the probability density function.
- Find the value of the constant k .
- Find $P(X > 2)$.

Solution



To find k , we can use the property that

$$\int_{\text{all } x} f(x) dx = 1 \text{ Note that}$$

$$(5-x)(x-2) = 7x - 10 - x^2.$$

$$\int_{\text{all } x} f(x) dx = k \int_1^3 (x-1) dx + k \int_3^5 7x - 10 - x^2 dx$$

$$\Rightarrow \frac{1}{k} = \left[\frac{1}{2} x^2 - x \right]_1^3 + \left[\frac{7}{2} x^2 - 10x - \frac{1}{3} x^3 \right]_3^5 = \left[\frac{3}{2} - \left(-\frac{1}{2}\right) \right] + \left[-\frac{25}{6} - \left(-\frac{15}{2}\right) \right] = \frac{16}{3} \therefore k = \frac{3}{16}$$

$$P(X > 2) = 1 - P(X \leq 2) = 1 - \frac{3}{16} \int_1^2 (x-1) dx = 1 - \frac{3}{16} \left[\frac{1}{2} x^2 - x \right]_1^2 = 1 - \frac{3}{16} \left[0 - \left(-\frac{1}{2}\right) \right] = \frac{29}{32}$$

Example 4 A continuous random variable X has a probability density function given by

$$f(x) = \begin{cases} 0.25, & 0 < x < 2 \\ 0.5x + c, & 2 < x < 3 \\ 0, & \text{elsewhere} \end{cases} \quad \text{Find } c \text{ hence compute } P(1 \leq X < 2.5).$$

Solution

$$\int_{\text{all } x} f(x)dx = 1 = \int_0^2 \frac{1}{4} dx + \int_2^3 \left(\frac{1}{2}x + c\right)dx = \frac{1}{2} + \frac{5}{4} + c \Rightarrow c = -\frac{3}{4}$$

$$P(1 \leq X < 2.5) = P(1 \leq X < 2) + P(2 \leq X < 2.5) = \int_1^2 \frac{1}{4} dx + \int_2^{2.5} \left(\frac{1}{2}x - \frac{3}{4}\right)dx = \frac{1}{4} + \left[\frac{x^2 - 3x}{4}\right]_2^{2.5} = \frac{1}{4} + \frac{3}{16} = \frac{7}{16}$$

Exercise 1.2

- 1) Suppose that the random variable X has p.d.f. given by $f(x) = \begin{cases} cx, & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$ Find the value of the constant c hence determine m so that $P(X \leq m) = \frac{1}{2}$
- 2) Let X be a continuous random variable with pdf $f(x) = \begin{cases} \frac{x}{5} + k, & 0 \leq x \leq 3 \\ 0, & \text{elsewhere} \end{cases}$ Find the value of the constant k hence compute $P(1 < X < 3)$
- 3) A continuous random variable Y has the pdf given by $f(y) = \begin{cases} k(1+y), & 4 \leq y \leq 7 \\ 0, & \text{elsewhere} \end{cases}$ Find the value of the constant k hence compute $P(Y < 5)$ and $P(5 < Y < 6)$
- 4) The continuous random variable X has probability density function $f(x) = \begin{cases} k(x-1)^2, & 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$ Find the value of the constant k hence compute $P(X < 1.5)$
- 5) A continuous random variable R is defined by the probability density function $f(r) = \begin{cases} k(5-r), & 0 \leq r \leq 2 \\ 0, & \text{otherwise} \end{cases}$
 - a) Sketch the p.d.f of R
 - b) Find k hence compute $P(1 \leq R \leq 3)$.
- 6) The life, T hours, of an electrical component is modelled by the probability density function $f(t) = \begin{cases} ke^{-0.001t}, & T \geq 1000 \\ 0, & \text{otherwise} \end{cases}$
 - a) Sketch the probability density function.
 - b) Find the value of the constant k .
 - c) Find $P(1500 \leq T \leq 2000)$.
- 7) A continuous r.v Y has probability density function $f(x) = \begin{cases} k(1-y^2), & -1 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$ Find the value of the constant k hence compute $P(Y < 0.5)$ and $P(-0.5 < Y < 0.5)$
- 8) Let X be a continuous r.v witha pdf $f(x) = \begin{cases} kx(1-x), & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$. Calculate $P\left(X - \frac{1}{2} > \frac{1}{4}\right)$
- 9) The pdf of a random variable X is given by $f(x) = \begin{cases} \frac{1}{6}, & 0 \leq x \leq 2 \\ \frac{1}{3}, & 2 \leq x \leq 4 \end{cases}$
Sketch the graph of $f(x)$ hence find $P(1 < x \leq 3)$
- 10) A continuous random variable X has the pdf given by $f(x) = \begin{cases} k(1+x)^2, & -2 \leq x \leq 0 \\ 4k, & 0 < x \leq \frac{4}{3} \\ 0, & \text{elsewhere} \end{cases}$ Find the value of the constant k hence compute $P(X > 1)$ and $P(-1 < X < 1)$

11) A continuous random variable X has the pdf given by $f(x) = \begin{cases} kx, & 0 \leq x \leq 2 \\ k(4-x), & 2 < x \leq 4 \\ 0, & \text{elsewhere} \end{cases}$ Find

the value of the constant k hence compute $P(X > 3)$ and $P(1 < X < 3)$

12) The length of time in minutes a customer queue in a post office is a random variable T with pdf

$$f(t) = \begin{cases} c(81-t^2), & 0 \leq t \leq 9 \\ 0, & \text{elsewhere} \end{cases}$$

- Find the value of the constant c and the cdf of T
- Find the probability that a customer will wait for more than 3 minutes
- A customer has been queuing for 3 minutes, find the probability that this customers will be queuing for at least 7 minutes
- Three customers are selected at random, find the probability that exactly 2 of them had to queue for longer than 3 min

1.4 Distribution Function of a Random Variables

Definition: For any random variable X , we define the **cumulative distribution function (CDF)**,

$$F(x) \text{ as } F(x) = P(X \leq x) = \begin{cases} \sum_{t=-\infty}^x P(T=t) & \text{If } X \text{ is discrete} \\ \int_{-\infty}^x f(t)dt & \text{If } X \text{ is continuous} \end{cases} \text{ for every } x.$$

Here t is introduced to facilitate summation /integration//

Properties of any cumulative distribution function

- $\lim_{x \rightarrow \infty} F(x) = 1$ and $\lim_{x \rightarrow -\infty} F(x) = 0$
- $F(x)$ is a non-decreasing function.
- $F(x)$ is a right continuous function of x . In other words $\lim_{t \rightarrow x} F(t) = F(x)$

Reminder If the cdf of X is $F(x)$ and the pdf is $f(x)$, then differentiate $F(x)$ to get $f(x)$, and integrate $f(x)$ to get $F(x)$;

Theorem: For any random variable X and real values $a < b$, $P(a \leq X \leq b) = F(b) - F(a)$

Example 1

Let X be a discrete random variable with pmf given by $f(x) = \begin{cases} \frac{1}{20}(1+x) & \text{for } x=1,2,3,4,5 \\ 0, & \text{elsewhere} \end{cases}$.

Determine the cdf of X hence compute $P(X > 3)$

Solution

$$F(x) = \sum_{t=-\infty}^x f(t) = \frac{1}{20} \sum_{t=1}^x (t+1) = \frac{1}{20} (2+3+\dots+x) = \frac{1}{20} \left\{ \frac{x}{2} [4+(x-1)] \right\} = \frac{x(x+3)}{40}$$

$$F(x) = \begin{cases} 0 & \text{for } x < 1 \\ \frac{x(x+3)}{40} & \text{for } x=1,2,3,4,5 \\ 1 & \text{for } x > 5 \end{cases} \quad \text{Recall for an AP } S_n = \frac{n}{2} [2a + (n-1)d]$$

$$P(X > 3) = 1 - P(X \leq 3) = 1 - \frac{3 \times 6}{40} = \frac{11}{20}$$

Example 2

Suppose X is a continuous random variable whose pdf f(x) is given by

$$f(x) = \begin{cases} \frac{1}{2}x, & 0 \leq x \leq 2 \\ 0, & \text{elsewhere} \end{cases}. \text{ Obtain the cdf of X hence compute } P(X > \frac{2}{3})$$

Solution

$$F(x) = \int_{-\infty}^x f(t)dt = \int_0^x \frac{1}{2}t dt = \frac{1}{4}t^2 \Big|_0^x = \frac{1}{4}x^2 \text{ thus } F(x) = \begin{cases} 0, & x < 0 \\ \frac{x^2}{4}, & 0 \leq x \leq 2 \\ 1, & x > 2 \end{cases}$$

$$P(X > \frac{2}{3}) = 1 - P(X \leq \frac{2}{3}) = 1 - \frac{1}{4}(\frac{2}{3})^2 = \frac{8}{9}$$

Example 3

A continuous random variable X has a probability density function given by

$$f(x) = \begin{cases} 0.25, & 0 < x < 2 \\ 0.5x + c, & 2 < x < 3 \\ 0, & \text{elsewhere} \end{cases} \text{ Find the cdf of X hence compute } P(1.5 \leq X < 2.5).$$

Solution

$$F(x) = \int_{-\infty}^x f(t)dt = \begin{cases} \int_0^x \frac{1}{4} dt = \frac{x}{4} \\ \int_2^x (\frac{1}{2}t - \frac{3}{4})dt + k = \frac{x^2 - 3x}{4} + \frac{1}{2} + k \end{cases}$$

Under the two levels, F(2) must be the same. (reason for introducing k)

$$\Rightarrow F(2) = \frac{1}{2} = k \quad \therefore F(x) = \begin{cases} 0 & \text{for } x < 0 \\ \frac{x}{4} & \text{for } 0 \leq x \leq 2 \\ \frac{x^2 - 3x}{4} + 1 & \text{for } 2 \leq x \leq 3 \\ 1, & \text{for } x > 3 \end{cases}$$

$$P(1.5 \leq X < 2.5) = F(2.5) - F(1.5) = \frac{2.5^2 - 3(2.5)}{4} + 1 - \frac{1.5}{4} = 0.3125$$

Exercise 1.3

1. The CDF of a discrete random variable X is given by $F(X) = \frac{x^3 + k}{40}, x = 1, 2, 3$

- a) Show that k=13 b) Find the pmf of x

2. The pdf of a continuous random variable X is given by $f(x) = \begin{cases} \frac{C}{\sqrt{x}}, & 0 \leq x \leq 4 \\ 0, & \text{elsewhere} \end{cases}$ Find the value of the constant C, the cdf of X and $P(X \geq 1)$

3. The pdf of a random variable X is given by $g(x) = \begin{cases} kx(1-x), & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$ Find the value of the constant k, the cdf of X and the value of x such that $G(x) = \frac{1}{2}$

4. A continuous random variable x has cumulative distribution function

$$F(x) = \begin{cases} 0, & x < -2 \\ \frac{1}{6}x + \frac{1}{3}, & -2 \leq x \leq 4 \\ 1, & x > 4 \end{cases} \text{ Find the}$$

probability density function f(x) of X.

5. A random variable X has p.d.f.

$$g(x) = \begin{cases} \frac{1}{6}(x^3 + 1), & 0 \leq x \leq 2 \\ 0, & \text{elsewhere} \end{cases}$$
 Find the c.d.f. of X hence obtain $P(X < 1)$.
6. A random variable X has pdf

$$f(x) = \begin{cases} \frac{1}{2}, & 0 < x < 1 \\ x - \frac{1}{2}, & 1 \leq x \leq k \text{ where } k \text{ is a constant.} \\ 0, & \text{otherwise} \end{cases}$$
 Sketch a graph of $f(x)$ hence find the value of k and the cdf of x hence compute $P(0.5 < x < 1.5)$
7. A continuous random variable X has pdf $f(x)$ given by

$$f(x) = \begin{cases} \frac{x^2}{45}, & 0 \leq x \leq 3 \\ 0.2, & 3 < x < 4 \\ \frac{10-x}{30}, & 4 \leq x \leq 10 \\ 0, & \text{elsewhere} \end{cases}$$
 Sketch $f(x)$. Also find the cdf of x and hence compute $p(X \leq 8)$
8. If the cdf of a random variable Y is given by $F(y) = 1 - \frac{9}{y^2}$ for $Y \geq 3$ and $F(x) = 0$ for $Y < 3$, find $P(X \leq 5)$, $P(X > 8)$ and the pdf of X
9. A random variable X has a cdf $F(X) = 2x^2 - x^4$ for $x \in (0,1)$. Compute $P(\frac{1}{4} < x \leq \frac{3}{4})$ and also find the pdf of x
10. Find the cdf of a random variable Y whose pdf is given by;
 a) $f(x) = \begin{cases} \frac{1}{3}, & 0 \leq x \leq 1 \\ \frac{1}{3}, & 2 \leq x \leq 4 \\ 0, & \text{elsewhere} \end{cases}$ b) $f(x) = \begin{cases} \frac{x}{2}, & 0 \leq x \leq 1 \\ \frac{1}{2}, & 1 \leq x \leq 2 \\ \frac{(3-x)}{2}, & 2 \leq x \leq 3 \\ 0, & \text{elsewhere} \end{cases}$
11. The continuous random variable x has probability density function $f(x)$ given by

$$f(x) = \begin{cases} k(x^2 - 2x + 2), & 0 < x \leq 3 \\ 3k, & 3 < x \leq 4 \\ 0, & \text{elsewhere} \end{cases}$$
 where k is a constant. Show that $k = \frac{1}{9}$
 a) Find the cumulative distribution function $F(x)$ and the mean of X .
 b) Show that the median of X lies between $x = 2.6$ and $x = 2.7$
12. The lifetime, X , in tens of hours, of a battery has a cumulative distribution function $F(x)$ given by

$$F(x) = \begin{cases} 0, & x < 1 \\ \frac{4}{9}(x^2 + 2x - 3), & 1 < x \leq 1.5 \\ 1, & x > 1.5 \end{cases}$$

 a) Find the median of X , giving your answer to 3 significant figures.
 b) Find, in full, the probability density function of the random variable X .
 c) Find $P(X > 1.2)$
13. The continuous random variable T represents the time in hours that students spend on homework. The cumulative distribution function of T is

$$F(t) = \begin{cases} 0, & t < 0 \\ k(2t^3 - t^4), & 0 < t \leq 1.5 \\ 1, & t > 1.5 \end{cases}$$
 where k is a positive constant.
 a) Show that $k = \frac{16}{27}$ hence and find the probability density function $f(t)$ of T .
 b) Find the proportion of students who spend more than 1 hour on homework.

- c) Show that $E(T) = 0.9$ and $F(E(T)) = 0.4752$.
- d) A student is selected at random. Given that the student spent more than the mean amount of time on homework, find the probability that this student spent more than 1 hour on homework.

1.5 Derived Random Variables

Give the pdf of a random variable say X , we can obtain the distribution of a second random variable say Y provided that we know some functional relationship between X and Y say

$Y = u(X)$. Eg $Y = 2X + 3$, $Y = X^3$, $Y = \sqrt{X} + c$ etc

For a 1-1 relationship between X and Y eg $Y = 2X + 3$, $f(x)$ and $g(y)$ yields exactly the same probabilities only the random variable and the set of values it can assume changes.

Example 1

Give the pmf of a random variable X as $f(x) = \begin{cases} \frac{x}{6} & \text{for } x=1,2,3 \\ 0, & \text{elsewhere} \end{cases}$ find the pmf of $Y = X^2$

Solution

The only values of Y with non-zero probabilities are $Y = 1$, $Y = 4$ and $Y = 9$. Now

$$P(Y = 1) = P(X^2 = 1) = P(X = 1) = \frac{1}{6} \quad P(Y = 4) = P(X^2 = 4) = P(X = 2) = \frac{1}{3} \quad \text{and}$$

$$P(Y = 9) = P(X^2 = 9) = P(X = 3) = \frac{1}{2}$$

In some cases several values of X will give rise to the same value of Y . The procedure is just the same as above but it is necessary to add the several probabilities that are associated with each value x that provides a unique value y .

Example 2 Given the pmf of a r.v X as $f(x) = \begin{cases} \frac{x+1}{15} & \text{for } x=0,1,2,3,4 \\ 0, & \text{elsewhere} \end{cases}$ find the pmf of

$$Y = (X - 2)^2$$

Solution

x	0	1	2	3	4
y	4	1	0	1	4

$$P(Y = 0) = P(X = 2) = \frac{1}{5} \quad P(Y = 1) = P(X = 1) + P(X = 3) = \frac{2}{15} + \frac{4}{15} = \frac{2}{5}$$

$$P(Y = 4) = P(X = 0) + P(X = 4) = \frac{1}{15} + \frac{5}{15} = \frac{2}{5} \quad \text{Therefore the pmf of } Y \text{ can be written as}$$

Y	0	1	4
$P(Y = y)$	$\frac{2}{5}$	$\frac{2}{5}$	$\frac{2}{5}$

Exercise 1.4

1. Suppose the pmf of a r.v X is given by $f(x) = \begin{cases} \frac{1}{6} & \text{for } x=1,2,3,4,5,6 \\ 0, & \text{elsewhere} \end{cases}$, Obtain the pmf of

$$Y = 2X^2 \text{ and } Z = X - 3$$

2. Let the pmf of a r.v X be given by $f(x) = \begin{cases} \frac{x^2+1}{18} & \text{for } x=0,1,2,3 \\ 0, & \text{elsewhere} \end{cases}$, determine the pmf of

$$Y = X^2 + 1$$

3. Suppose the pmf of a r.v X is given by $f(x) = \begin{cases} \frac{x}{10} & \text{for } x = 0, 1, 2, 3, 4 \\ 0, & \text{elsewhere} \end{cases}$, Obtain the pmf of $Y = |X - 2|$
 4. Let the pmf of a r.v X be given by $f(x) = \begin{cases} \left(\frac{1}{2}\right)^x & \text{for } x = 1, 2, 3, \dots \\ 0, & \text{elsewhere} \end{cases}$, determine the pmf of $Y = \begin{cases} 1 & \text{if } x \text{ is odd} \\ 0 & \text{if } x \text{ is even} \end{cases}$
 5. Suppose the r.v X has a pmf given by $f(x) = \begin{cases} \frac{1}{6} \left(\frac{5}{6}\right)^x & \text{for } x = 0, 1, 2, 3, \dots \\ 0, & \text{elsewhere} \end{cases}$, Obtain the pmf of $Y = X - 1$
 6. Let X be a discrete r.v with pmf as tabulated below. Find the pmf of $Y = (X + 2)^2$
- | | | | | | |
|----------|---------------|---------------|---------------|---------------|---------------|
| X | -2 | -1 | 0 | 1 | 2 |
| $P(X=x)$ | $\frac{1}{4}$ | $\frac{1}{8}$ | $\frac{1}{8}$ | $\frac{1}{4}$ | $\frac{1}{4}$ |

1.6 Change of Variable Technique

.Let X be a r.v with pdf $f(x)$ and let Y be a function of X , then the pdf of Y , $g(y) = f(x) \left| \frac{dx}{dy} \right|$

NB If $F(x)$ is the cdf of a r.v X , then a r.v $Y = F(x)$ has a uniform distribution over $[0, 1]$

Example 1

A continuous r.v X has a pdf given by $f(x) = \begin{cases} 5x^4, & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$. Determine the pdf of $Y = x^3$

Solution

$$Y = x^3 \Rightarrow X = Y^{1/3} \Rightarrow \frac{dx}{dy} = \frac{1}{3} Y^{-2/3} \quad g(y) = f(x) \left| \frac{dx}{dy} \right| = 5 \left(y^{1/3} \right)^4 \times \frac{1}{3} y^{-2/3} = \begin{cases} \frac{5}{3} y^{2/3}, & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Example 2 A r.v X has pdf $f(x) = \begin{cases} 24x^2, & 0 \leq x \leq \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$ determine the pdf of $Y = 8X^3$

Solution

$$Y = 8x^3 \Rightarrow X = \frac{1}{2} Y^{1/3} \Rightarrow \frac{dx}{dy} = \frac{1}{6} y^{-2/3} \quad g(y) = f(x) \left| \frac{dx}{dy} \right| = 24 \left(\frac{1}{2} y^{1/3} \right)^2 \times \frac{1}{6} y^{-2/3} = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

NB $Y = 8X^3$ is the cdf of X

Exercise 1.5

1. For a r.v X with $f(x) = \begin{cases} 5x^4, & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$, determine the pdf of $Y = 2 \ln x$ and its range.
2. A r.v X has pdf $f(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$ determine the pdf of $Y = X^4$

3. Let X be a continuous random variable with density function $f(x) = \frac{1}{x^2}$ for $x \geq 2$ and $f(x) = 0$ otherwise. Determine the density function of $Y = \frac{1}{X-1}$.
4. The probability density function of X is given by $f(x) = \begin{cases} \frac{2}{\pi(x^2 + 4)} & -\infty \leq X \leq \infty \\ 0 & \text{otherwise} \end{cases}$ Obtain the probability density function of $Y = \tan^{-1}\left(\frac{x}{2}\right)$
5. Which transformation will change a r.v X with pdf is as below to a uniform R.V Y whose range is $0 \leq x \leq 1$
 - a) $f(x) = \begin{cases} 2e^{-2x}, & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$
 - b) $f(x) = \begin{cases} \frac{1}{2}(x-3), & 3 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$
6. Suppose that X has probability density function $f(x) = \begin{cases} \frac{\theta}{x \ln \theta} & \text{for } x > 1 \\ 0, & \text{elsewhere} \end{cases}$ If $U = X^2$, what is the probability density function $f(u)$ of U ?

1.7 Expectation and Variance of a Random Variable

1.7.1 Expected Values

One of the most important things we'd like to know about a random variable is: what value does it take on average? What is the average price of a computer? What is the average value of a number that rolls on a die? The value is found as the average of all possible values, weighted by how often they occur (i.e. probability)

Definition: Let X be a random variable with probability distribution $p(X = x)$. Then the **expected value** of X , denoted $E(X)$ or μ , is given by;

$$E(X) = \mu = \begin{cases} \sum_{x=-\infty}^{\infty} xp(X = x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} xp(X = x)dx & \text{if } X \text{ is continuous} \end{cases}.$$

Theorem: Let X be a r.v. with probability distribution $p(X=x)$ and let $g(x)$ be a real-valued function of X . ie $g: \mathbb{R} \rightarrow \mathbb{R}$, then the expected value of $g(x)$ is given by

$$E[g(x)] = \begin{cases} \sum_{x=-\infty}^{\infty} g(x)p(X = x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} g(x)p(X = x)dx & \text{if } X \text{ is continuous} \end{cases}.$$

Theorem: Let X be a r.v. with probability distribution $p(X = x)$. Then

- (i) $E(c) = c$, where c is an arbitrary constant.
- (ii) $E[ax + b] = a\mu + b$ where a and b are arbitrary constants
- (iii) $E[kg(x)] = kE[g(x)]$ where $g(x)$ is a function of X
- (iv) $E[ag_1(x) \pm bg_2(x)] = aE[g_1(x)] \pm bE[g_2(x)]$ and in general $E\left[\sum_{i=1}^n c_i g_i(x)\right] = \sum_{i=1}^n c_i E[g_i(x)]$

are functions of X . This property of expectation is called *linearity property*

Proof

We will sketch the proof using a continuous random variable since the proof using the discrete random variable is similar and was also discussed in probability and statistics I.

- (i) $E[c] = \int_{All\ x} cP(X=x)dx = c \int_{All\ x} P(X=x)dx = c(1) = c$
- (ii) $E[ax+b] = \int_{All\ x} (ax+b)P(X=x)dx = \int_{All\ x} axP(X=x)dx + \int_{All\ x} bP(X=x)dx$
 $= a \int_{All\ x} xP(X=x)dx + b \int_{All\ x} P(X=x)dx = a\mu + b$
- (iii) $E[kg(x)] = \int_{All\ x} kg(x)P(X=x)dx = k \int_{All\ x} g(x)P(X=x)dx = kE[g(x)]$
- (iv) $E[ag_1(x) \pm bg_2(x)] = \int_{All\ x} [ag_1(x) \pm bg_2(x)]P(X=x)dx = \int_{All\ x} ag_1(x)P(X=x)dx \pm \int_{All\ x} bg_2(x)P(X=x)dx$
 $= E[ag_1(x)] \pm E[bg_2(x)] = aE[g_1(x)] \pm bE[g_2(x)]$ ie from part iii

1.7.2 Variance and Standard Deviation

Definition: Let X be a r.v with mean $E(X) = \mu$, the **variance** of X , denoted σ^2 or $\text{Var}(X)$, is given by $\text{Var}(X) = \sigma^2 = E(X - \mu)^2$. The units for variance are square units. The quantity that has the correct units is **standard deviation**, denoted σ . It's actually the positive square root of $\text{Var}(X)$.

$$\sigma = \sqrt{\text{Var}(X)} = \sqrt{E(X - \mu)^2}.$$

Theorem: $\text{Var}(X) = E(X - \mu)^2 = E(X)^2 - \mu^2$

Proof:

$$\text{Var}(X) = E(X - \mu)^2 = E(X^2 - 2X\mu + \mu^2) = E(X)^2 - 2\mu E(X) + \mu^2 = E(X)^2 - \mu^2$$

Theorem: $\text{Var}(aX + b) = a^2 \text{var}(X)$

Proof:

Recall that $E[aX + b] = a\mu + b$ therefore

$$\text{Var}(aX + b) = E[(aX + b) - (a\mu + b)]^2 = E[a(X - \mu)]^2 = E[a^2(X - \mu)^2] = a^2 E[(X - \mu)^2] = a^2 \text{var}(X)$$

Remark

(i) The expected value of X always lies between the smallest and largest values of X .

(ii) In computations, bear in mind that variance cannot be negative!

Example 1

Given a probability distribution of X as below, find the mean and standard deviation of X .

X	0	1	2	3
$P(X=x)$	1/8	1/4	3/8	1/4

Solution

X	0	1	2	3	total
$p(X=x)$	1/8	1/4	3/8	1/4	1
$xp(X=x)$	0	1/4	3/4	3/4	7/4
$x^2 p(X=x)$	0	1/4	3/2	9/4	4

$$E(X) = \mu = \sum_{x=0}^3 xp(X=x) = 1.75 \text{ and}$$

standard deviation

$$\sigma = \sqrt{E(X^2) - \mu^2} = \sqrt{4 - 1.75^2} \approx 0.968246$$

Example 2 The probability distribution of a r.v X is as shown below, find the mean and standard deviation of; a) X b) $Y = 12X + 6$.

X	0	1	2
$P(X=x)$	1/6	1/2	1/3

Solution

X	0	1	2	total
$p(X = x)$	1/6	1/2	1/3	1
$xp(X = x)$	0	1/2	2/3	7/6
$x^2 p(X = x)$	0	1/2	4/3	11/6

$$E(X) = \mu = \sum_{x=0}^2 xp(X = x) = 7/6 \quad \text{and}$$

$$E(X^2) = \sum_{x=0}^2 x^2 p(X = x) = 11/6$$

$$\text{Standard deviation } \sigma = \sqrt{E(X^2) - \mu^2} = \sqrt{11/6 - (7/6)^2} = \sqrt{17/6} \approx 1.6833$$

$$\text{Now } E(Y) = 12E(X) + 6 = 12(7/6) + 6 = 20$$

$$\text{Var}(Y) = \text{Var}(12X + 6) = 12^2 \times \text{Var}(X) = 144 \times \sqrt{17/6} \approx 242.38812$$

Example 3 A continuous r.v X has a pdf given by $f(x) = \begin{cases} \frac{1}{2}x, & 0 \leq x \leq 2 \\ 0, & \text{elsewhere} \end{cases}$, find the mean

and standard of X

Solution

$$E(x) = \int_{-\infty}^{\infty} xf(x)dx = \int_0^2 \frac{1}{2}x^2 dx = \left[\frac{1}{6}x^3 \right]_0^2 = \frac{4}{3} \quad \text{and} \quad E(x^2) = \int_{-\infty}^{\infty} x^2 f(x)dx = \int_0^2 \frac{1}{2}x^3 dx = \left[\frac{1}{8}x^4 \right]_0^2 = 2$$

$$\text{Standard deviation } \sigma = \sqrt{E(X^2) - \mu^2} = \sqrt{2 - (4/3)^2} = \frac{\sqrt{2}}{3}$$

Exercise 1.6

1. Suppose X has a probability mass function given by the table below

X	2	3	4	5	6
$P(X=x)$	0.01	0.25	0.4	0.3	0.04

Find the mean and variance of; X

2. Suppose X has a probability mass function given by the table below

X	11	12	13	14	15
$P(X=x)$	0.4	0.2	0.2	0.1	0.1

Find the mean and variance of; X

3. Let X be a random variable with $P(X = 1) = 0.2$, $P(X = 2) = 0.3$, and $P(X = 3) = 0.5$. What is the expected value and standard deviation of; a) X b) $Y = 5X - 10$?
4. A random variable W has the probability distribution shown below,

W	0	1	2	3
$P(W=w)$	2d	0.3	d	0.1

Find the values of the constant d hence determine the mean and variance of W . Also find the mean and variance of $Y = 10X + 25$

5. A random variable X has the probability distribution shown below,

X	1	2	3	4	5
$P(X=x)$	7c	5c	4c	3c	c

Find the values of the constant c hence determine the mean and variance of X .

6. The random variable Z has the probability distribution shown below,

Z	2	3	5	7	11
-----	---	---	---	---	----

$P(Z=z)$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{4}$	x	y
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If $E(Z) = 4\frac{2}{3}$, find the values of x and y hence determine the variance of Z

7. A discrete random variable M has the probability distribution $f(m) = \begin{cases} \frac{m}{36}, & m=1,2,3,\dots,8 \\ 0, & \text{elsewhere} \end{cases}$,
find the mean and variance of M
8. For a discrete random variable Y the probability distribution is $f(y) = \begin{cases} \frac{5-y}{10}, & y=1,2,3,4 \\ 0, & \text{elsewhere} \end{cases}$,
calculate $E(Y)$ and $\text{var}(Y)$
9. Suppose X has a pmf given by $f(x) = \begin{cases} kx & \text{for } x=1,2,3,4 \\ 0, & \text{elsewhere} \end{cases}$, find the value of the constant k
hence obtain the mean and variance of X
10. A discrete random variable X has a probability function $P(X = x) = \begin{cases} kx, & x=2, 4, 6 \\ k(x-2), & x=8 \\ 0, & \text{elsewhere} \end{cases}$
 - a) Show that $k = \frac{1}{18}$ hence find $E(X)$ and $E(X^2)$
 - b) Calculate $\text{var}(3-4X)$, give your answer to 3 significant figures.
11. A biased die with six faces is rolled. The discrete random variable X represents the score of the uppermost face. The probability distribution of X is shown in the table below

X	1	2	3	4	5	6
$P(X=x)$	A	a	a	b	B	0.3

 - a) Given that $E(X) = 4.2$, find the values of a and b .
 - b) Show that $E(X^2) = 20.4$
 - c) Find $\text{var}(5-3x)$
12. A discrete random variable X has probability function $P(X = x) = \begin{cases} k(1-x)^2, & x=-1,0,1,2 \\ 0, & \text{elsewhere} \end{cases}$
 - a) Show that $k = \frac{1}{6}$
 - b) Find $E(X)$
 - c) Show that $E(X^2) = \frac{4}{3}$
 - d) Find $\text{var}(1-3x)$
13. A team of 3 is to be chosen from 4 girl and 6 boys. If X is the number of girls in the team, find the probability distribution of X hence determine the mean and variance of X
14. A fair six sided die has; '1' on one face, '2' on two of its faces and '3' on the remaining three faces. The die is rolled twice. If T is the total score write down the probability distribution of T hence determine; a) the probability that T is more than 4. b) the mean and variance of T
15. The pdf of a continuous r.v R is given by $f(r) = \begin{cases} kr & \text{for } 0 \leq r \leq 4 \\ 0, & \text{elsewhere} \end{cases}$, (a) Determine c . hence
Compute $P(10 \leq r \leq 2)$, $E(X)$ and $\text{Var}(X)$.
16. A continuous r.v M has the pdf given by $f(m) = \begin{cases} k(1-\frac{m}{10}) & \text{for } M \leq 10 \\ 0, & \text{elsewhere} \end{cases}$, find the value of
the constant k , the mean and the variance of X

- 17 A continuous r.v X has the pdf given by $f(x) = \begin{cases} k(1-x) & \text{for } 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$, find the value of the constant k . Also find the mean and the variance of X
- 18 The lifetime of new bus engines, T years, has continuous pdf $f(t) = \begin{cases} \frac{d}{t^2} & \text{if } T \geq 1 \\ 0, & \text{if } T < 1 \end{cases}$ find the value of the constant d hence determine the mean and standard deviation of T
- 19 An archer shoots an arrow at a target. The distance of the arrow from the centre of the target is a random variable X whose p.d.f. is given by $f(x) = \begin{cases} k(3+2x-x^2) & \text{if } x \leq 3 \\ 0, & \text{if } x > 3 \end{cases}$ find the value of the constant k . Also find the mean and standard deviation of X
- 20 The random variable Y has probability density function $f(y)$ given by $f(y) = \begin{cases} ky(a-y), & 0 \leq y \leq 3 \\ 0, & \text{elsewhere} \end{cases}$ where k and a are positive constants.
- Explain why $a \geq 3$ and then show that $k = \frac{2}{9(a-2)}$
 - Given that $E(Y) = 1.75$, show that $a = 4$ and write down the value of k .
 - For these values of a and k , sketch the probability density function,
 - Write down the mode of Y .
- 21 A continuous random variable x has the following pdf $f(x) = \begin{cases} a+bx, & 0 \leq x \leq 5 \\ 0, & \text{otherwise} \end{cases}$ where a and b are constants. Show that $10a + 25b = 2$
- Given $E(X) = \frac{35}{12}$, find a second equation in a and b hence find the values of a and b .
 - Find the median of X
- 22 The queuing time X minutes of a customer at a till of a supermarket has a pdf $f(x) = \begin{cases} \frac{3}{32}x(k-x), & 0 \leq x \leq k \\ 0, & \text{otherwise} \end{cases}$
- Show that $k = 4$. Also find $E(x)$ and $\text{var}(x)$
 - Find the probability that a randomly chosen customers queuing time will differ from the mean by at least half a minute
- 23 The probability density function $f(x)$ can be written in the following form.
- $$f(x) = \begin{cases} ax, & 0 \leq x \leq 2 \\ b-ax, & 2 \leq x \leq 4 \\ 0, & \text{otherwise} \end{cases}$$
- Find the values of the constants a and b .
 - Show that σ , the standard deviation of X , is 0.816 to 3 decimal places.
 - Find the lower quartile of X .
 - State, giving a reason, whether $P(2 - \sigma < X < 2 + \sigma)$ is more or less than 0.5
- 24 A continuous r.v X has the pdf given by $f(x) = \begin{cases} k(1+x), & -1 \leq x < 0 \\ 2k(1-x), & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$, find the value of the constant k . Also find the mean and the variance of X

- 25 A continuous r.v X has the pdf given by $f(x) = \begin{cases} e^{-x} & \text{for } x > 0 \\ 0, & \text{elsewhere} \end{cases}$, find the mean and standard deviation of; a) X b) $Y = e^{\frac{3}{4}x}$

1.8 Mode, Median, Quartiles and Percentiles

Another measure commonly used to summarize random variables are the mode and median;

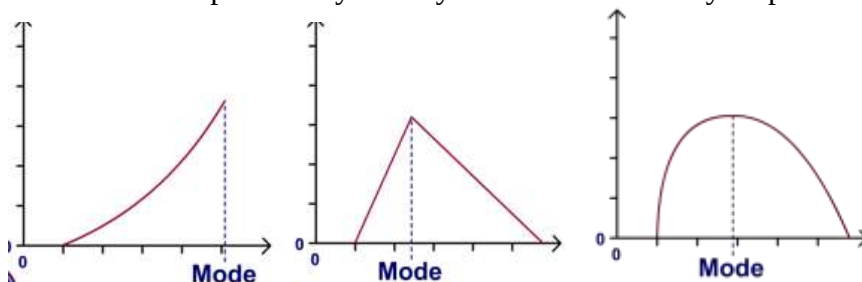
Mode

Mode is the value of x that maximizes the pdf. That is the value of x for which $f'(x) = 0$.

Suppose the p.d.f of a random variable X is defined by the function $f(x)$ for $a \leq x \leq b$.

The **mode** of X is an x value that produces the largest value for $f(x)$ in the interval $a \leq x \leq b$.

A sketch of the probability density function can be very helpful when determining the mode.



For the 3rd figure, mode can be found using differentiation.

Median

The **median**, m , of a random variable X is defined to be the value such that “half of the distribution lies to the left of m and half to the right”. More formally, m should satisfy $F_X(m) = P(X \leq m) = 0.5$ where F is the cumulative distribution function of X .

Note: If there is a value m such that the graph of $y = f(x)$ is symmetric about $x = m$, then both the expected value and the median of X are equal to m .

The lower quartile Q_1 and the upper quartile Q_3 are similarly defined by

$$F_X(Q_1) = 0.25 \text{ and } F_X(Q_3) = 0.75$$

Thus, the probability that X lies between Q_1 and Q_3 is $0.75 - 0.25 = 0.5$, so the quartiles give an estimate of how spread-out the distribution is. More generally, we define the n^{th} percentile of X to be the value of x_n such that $F_X(x_n) = 0.01n$ or $\frac{n}{100}$, that is, the probability that X is smaller than x_n is $n\%$.

Example A random variable X has the pdf given by $f(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$ State the mode

hence find the lower, middle and upper quartiles.

Solution

On the interval $0 \leq x \leq 1$, $f(x)$ is maximized at $x = 1$ so the mode is 1

On the interval $0 \leq x \leq 1$, the cdf of X is given by $F(x) = x^2$ thus

a) At lower quartile Q_1 , $F(Q_1) = Q_1^2 = 0.25 \Rightarrow Q_1 = \sqrt{0.25} = 0.5$

b) At median m , $F(m) = m^2 = 0.5 \Rightarrow m = \sqrt{0.5} = \frac{1}{\sqrt{2}}$

c) At upper quartile u $F(Q_3) = Q_3^2 = 0.75 \Rightarrow Q_3 = \sqrt{0.75} = \frac{\sqrt{3}}{2}$

Qn Find the 64th percentile of the pdf in the above example.

Exercise 1.7

1) A continuous r.v X has the pdf given by $f(x) = \begin{cases} k(1-x) & \text{for } 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$, find the mode.

2) A random variable X has p.d.f. $f(x) = \begin{cases} x^2(x-2) & 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$ Find the mode of X .

3) Let X be a continuous random variable with density function $f(x) = \begin{cases} \frac{1}{2}e^{-\frac{x}{2}} & \text{for } x > 0 \\ 0, & \text{elsewhere} \end{cases}$.

Determine the 25th percentile of the distribution of X .

4) A random variable X is defined by the cumulative distribution function

$$F(x) = \begin{cases} 0, & x < 2 \\ k(x^2 + x - 6), & 2 \leq x \leq 5 \\ 1, & x > 5 \end{cases}$$

- Find the value of the constant k hence work out $P(3 \leq X \leq 4)$.
- Obtain and sketch the probability density function.
- State the mode. Also find median value and the interquartile range.

5) A random variable X has p.d.f. $f(x)$, where $f(x) = \begin{cases} k(x^2 - 2x + 3) & 1 \leq x \leq 2 \\ k(2 - \frac{1}{2}x) & 2 \leq x \leq 4 \\ 0, & \text{otherwise} \end{cases}$

- Find the C.D.F of X and verify that the lower quartile is at $x = 2$.
- Obtain the mode and the median value of X .

2. PROBABILITY DISTRIBUTION

2.1 Discrete Distribution

Among the discrete distributions that we will discuss in this topic includes the Bernoulli, binomial, Poisson, geometric and hyper-geometric

Definition: A **Bernoulli trial** is a random experiment in which there are only two possible outcomes - success and failure. Eg

- Tossing a coin and considering heads as success and tails as failure.
- Checking items from a production line: success = not defective, failure = defective.
- Phoning a call centre: success = operator free; failure = no operator free.

A Bernoulli random variable X takes the values 0 and 1 and $P(X = 1) = p$ and

$$P(X = 0) = 1 - p$$

Definition: A r.v X is said to be a real Bernoulli distribution if it's pmf is given by;

$$P(X = x) = \begin{cases} p^x(1-p)^{1-x} & \text{for } x = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

We abbreviate this as $X \sim B(p)$ ie p is the only parameter here. It can be easily checked that the mean and variance of a Bernoulli random variable are $\mu = p$ and $\sigma^2 = p(1-p)$

2.1.1 Binomial Distribution

Consider a sequence of n independent, Bernoulli trials, with each trial having two possible outcomes, *success* or *failure*. Let p be the probability of a success for any single trial. Let X

denote the number of successes on n trials. The random variable X is said to have a **binomial distribution** and has probability mass function

$$P(X = x) = {}_n C_r \times p^x (1-p)^{n-x} \text{ for } x = 0, 1, 2, \dots, n$$

We abbreviate this as $X \sim \text{Bin}(n, p)$ read as “ X follows a binomial distribution with parameters n and p ”. ${}_n C_r$ Counts the number of outcomes that include exactly x successes and $n-x$ failures.

The mean and variance of a Binomial random variable are respectively given by;

$$\mu = np \text{ and } \sigma^2 = np(1-p)$$

Let's check to make sure that if X has a binomial distribution, then $\sum_{x=0}^n P(X = x) = 1$. We will need the binomial expansion for any polynomial:

$$(p+q)^n = \sum_{x=0}^n {}_n C_r \times p^x q^{n-x} \text{ therefore } \sum_{x=0}^n {}_n C_x \times p^x (1-p)^{n-x} = [p + (1-p)]^n = 1^n = 1$$

Example 1

A biased coin is tossed 6 times. The probability of heads on any toss is 0:3. Let X denote the number of heads that come up. Calculate: (i) $P(X = 2)$ (ii) $P(X = 3)$ (iii) $P(1 < X < 5)$

Solution

If we call heads a success then X has a binomial distribution with parameters $n=6$ and $p=0:3$.

$$(i) P(X = 2) = {}_6 C_2 \times (0.3)^2 (0.7)^4 = 0.324135$$

$$(ii) P(X = 3) = {}_6 C_3 \times (0.3)^3 (0.7)^3 = 0.18522$$

$$(iii) P(1 < X \leq 5) = P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5) \\ = 0.324 + 0.185 + 0.059 + 0.01 = 0.578$$

Example 2 A quality control engineer is in charge of testing whether or not 90% of the DVD players produced by his company conform to specifications. To do this, the engineer randomly selects a batch of 12 DVD players from each day's production. The day's production is acceptable provided no more than 1 DVD player fails to meet specifications'. Otherwise, the entire day's production has to be tested.

- What is the probability that the engineer incorrectly passes a day's production as acceptable if only 80% of the day's DVD players actually conform to specification?
- What is the probability that the engineer unnecessarily requires the entire day's production to be tested if in fact 90% of the DVD players conform to specifications?

Solution

Let X denote the number of DVD players in the sample that fail to meet specifications.

- In part a we want $P(X \leq 1)$ with binomial parameters $n = 12$ and $p = 0.2$

$$P(X \leq 1) = P(X = 0) + P(X = 1) = {}_{12} C_0 \times (0.2)^0 (0.8)^{12} + {}_{12} C_1 \times (0.2)^1 (0.8)^{11} \\ = 0.069 + 0.206 = 0.275$$

- In part b we require $P(X > 1) = 1 - P(X \leq 1)$ with parameters $n = 12$ and $p = 0.1$.

$$P(X \leq 1) = P(X = 0) + P(X = 1) = {}_{12} C_0 \times (0.1)^0 (0.9)^{12} + {}_{12} C_1 \times (0.1)^1 (0.9)^{11} = 0.659$$

$$\text{So } P(X > 1) = 0.341$$

Example 3 Bits are sent over a communications channel in packets of 12. If the probability of a bit being corrupted over this channel is 0.1 and such errors are independent, what is the probability that no more than 2 bits in a packet are corrupted?